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CIS 5660 Final Project Pitch

Strand Type Project

This project will attempt to recreate Death Stranding environment generation in my personal Vulkan engine. The project will be a study of 2 to 3 areas from the game series and attempt to recreate the terrain and foliage utilizing procedural generation. There are a lot of resources on terrain generation and foliage rendering, including some from the Death Stranding engine (Decima)'s own team, however, due to the complexity of their systems, I will likely have to look elsewhere in order to create my own system that works for the current engine infrastructure. The engine will likely have to be slightly modified, the details of which are adding a rendergraph to allow for occlusion culling, terrain generation pass, and a mesh shader pass for foliage rendering.

Features:

- Procedurally generated terrain
- Procedurally generated grass using mesh shaders
- Engine features upgrades to make it efficient
    - Frustum culling
    - Occlusion culling

Final Product:

- Demo of engine showing off procedural terrain/foliage

Milestones:

- Week 1: Research + engine infrastructure upgrades
- Week 2: Terrain generation
- Week 3: Foliage asset generation/placement

References:

- The entire body of text on terrain generation (video is a nice starting point)
- Death Stranding Frame Breakdown
- CIS 5650 Vulkan Grass Rendering
- Procedural Grass in 'Ghost of Tsushima'
- GPU-Based Procedural Placement in Horizon Zero Dawn
- Procedural World Generation of Far Cry 5